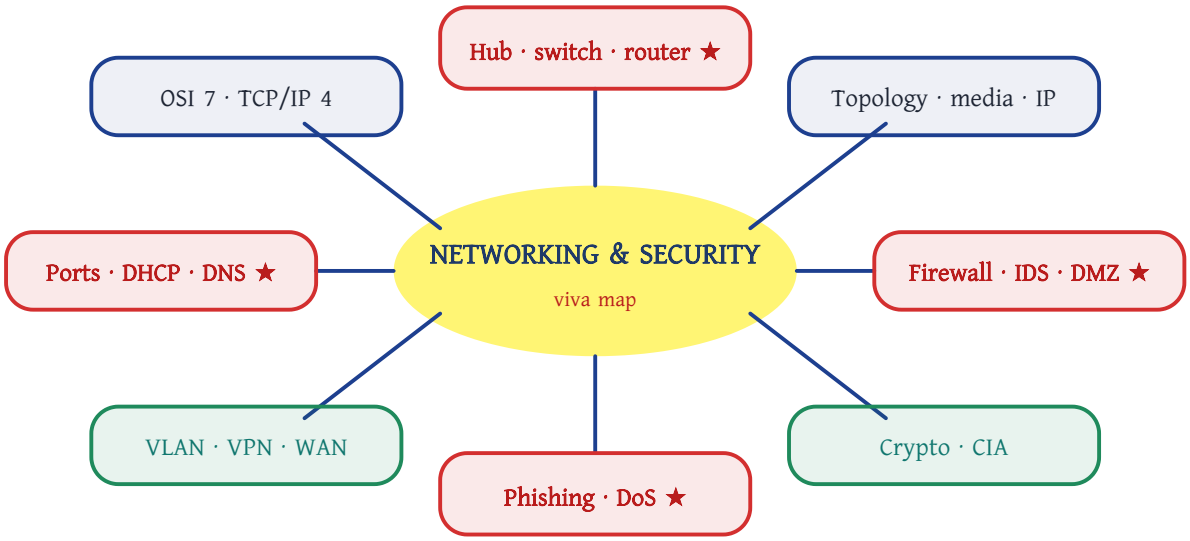
Viva Questions Solved: Networking & Security

122 viva questions from bank IT, BPSC and NTRCA vivas, solved in 12 sheets



↑ what the boards ask: red = asked most often

Papers covered

- Bank IT vivas: 101 questions compiled from candidates' viva experiences [B]
- BPSC viva 2023, Assistant Maintenance Engineer [P23] · BPSC viva, Assistant Programmer, ICT Division [P]
- NTRCA 17th registration viva 2023, Lecturer (ICT) [N23] · two untagged security questions

Most asked
① devices + layer (17×)
② attacks, malware (14×)
③ protocols, ports (12×)
④ firewall, IDS, DMZ (11×)
⑤ encryption (10×)
⑥ OSI / TCP/IP (9×)

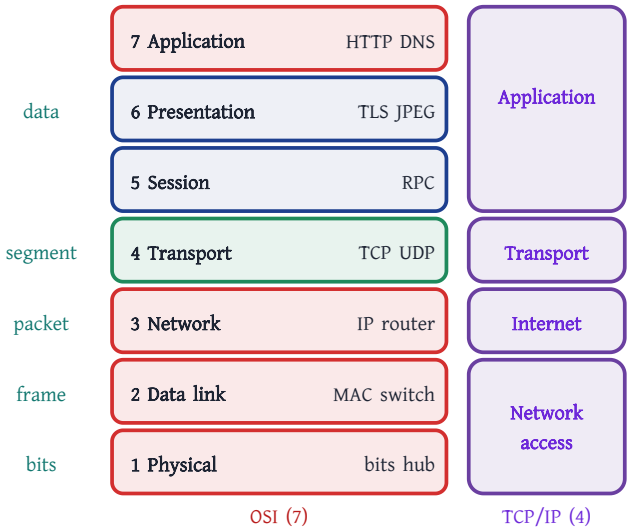
Legend: **yellow** = key term · **red circle** = must-memorise · **purple** = memory trick · **red box** = trap · **green** = model answer · [B] = where asked

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1 OSI & TCP/IP Reference Models

Recap: **OSI**: ISO's **7-layer reference model** (ISO 7498, 1984), a framework, not a protocol. **TCP/IP**: the **4-layer suite** the Internet runs. Each layer adds its header on the way down.



OSI ↔ TCP/IP, PDU per layer

Memory trick: 1 → 7: "**P**lease **D**o **N**ot **T**hrow **S**ausage **P**izza **A**way". TCP/IP top-down: **ATIN**.

- **How is ISO applied in networking?** [B] ISO published OSI (ISO/IEC 7498-1) so multi-vendor equipment interoperates; banks also follow ISO 8583 (card messages) and ISO 27001.
- **What is OSI? How many layers? Name them.** [P23 · N23 · B] ISO's Open Systems Interconnection model, **7 layers**, bottom → top: Physical, Data link, Network, Transport, Session, Presentation, Application.
- **TCP/IP: how many layers?** [N23] **4**: Application, Transport, Internet, Network access (link). Some books use 5, splitting the last into data link + physical.
- **OSI vs TCP/IP?** [B] OSI: 7 layers, model before protocols, generic, for teaching. TCP/IP: 4 layers, protocols first, the Internet's working standard; session + presentation sit inside its application layer.
- **What is framing? Which layer?** [B · B] The **data link layer** cuts the bit stream into frames: header (MAC addresses) + payload + trailer (FCS/CRC), with frame boundaries marked.

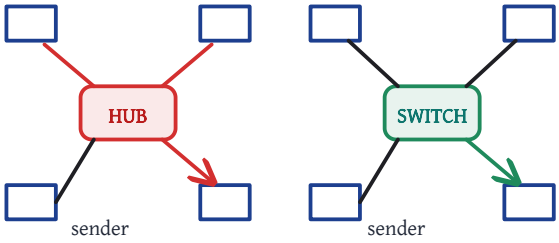
Model answer ✍️: Network for one office room: which layers, which tools? [P23]

1. **L1:** star layout; Cat6 UTP, RJ-45, crimper, **cable tester**, NICs, switch, Wi-Fi AP.
2. **L2:** managed switch (VLANs, port security), WPA2/WPA3 Wi-Fi; check link LEDs.
3. **L3:** IP plan, DHCP, default gateway, DNS; test with *ipconfig*, *ping*.
4. **L4–7:** firewall port rules, file/printer sharing, antivirus; test with *nslookup*.

Exam trap: "Lower 3 = hardware, upper 3 = software" is loose: IP (L3) is software in every host. Say **1–3 network-support, 4 transport, 5–7 user-support**. TCP/IP's last layer is network *access*; OSI's network layer = TCP/IP's Internet layer.

2 Network Devices & Their OSI Layers

	Repeater / Hub	Switch	Router	Gateway
OSI layer [B]	1	2	3	any, mostly 4–7
Decides by	nothing	MAC table	IP routing table	protocol mapping
Sends to	all ports	one port	best path	other protocol
Collision domains	one, shared	one per port	one per port	–
Broadcast domains	one	one per VLAN	one per interface	–



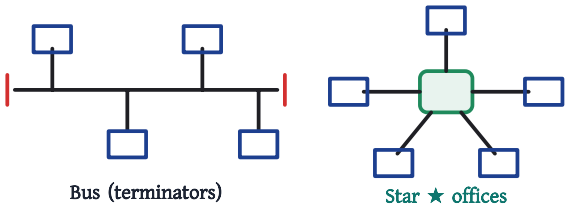
hub floods · switch sends to one

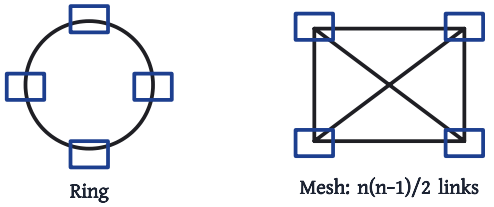
> **Memory trick:** **1-2-3**: the **hub** shouts to everyone, the **switch** reads the MAC on the envelope, the **router** reads the IP "city". The **gateway** is the interpreter.

- **What is a router? Main function?** [N23 · B] An L3 device that joins **different networks** and forwards each packet by destination IP over the best path in its routing table; at the edge it also does NAT, DHCP, ACLs.
- **Which layer: switch? router?** [B · B] Switch L2 (an L3 switch also routes between VLANs); router L3, as it forwards by IP.
- **What is a switch? Can it link many PCs?** [B · B] Yes: a multiport bridge for one LAN; it learns MACs into its CAM table and sends each frame only to the destination port.
- **What is a repeater?** [B] L1: **regenerates** a weak signal to extend a segment (past 100 m of UTP). A hub is a multiport repeater.
- **Gateway? Gateway vs router?** [N23 ×2] A gateway links networks with **different protocols** and translates (VoIP, mail, payment gateways). A router links same-protocol (IP) networks and only forwards.
- **Hub vs switch; is the hub obsolete?** [N23 · B] Hub: repeats to all ports, one collision domain, half-duplex, easy sniffing. Switch: filters by MAC, full-duplex. So yes.
- **Network with only a router? only a hub?** [B · B] Router: for a few PCs yes, a SOHO router has a built-in switch + Wi-Fi; for more, add a switch. Hub: works, but slow and insecure.
- **USB-based router?** [B] Shares a USB 3G/4G dongle's link over Wi-Fi/LAN: a branch backup link.

> **Exam trap:** The "default gateway" in IP settings is just your **router's interface**, not a protocol translator.

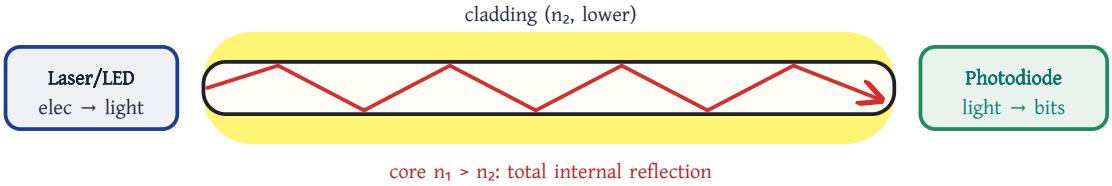
3 Topologies & Network Types





- **What is topology? Its types?** [N23 · B ×2] The physical or logical layout of nodes and links: **bus, star, ring, mesh, tree, hybrid.**
- **Bus and star?** [N23] Bus: all nodes tap one terminated cable; a frame reaches all, only the addressee keeps it; one break stops all. Star: each node cabled to a central switch; a cable fault isolates one node, but the switch is a single point of failure.
- **Which topology do offices use?** [B] **Star** (a tree of switches): easy to add users, find faults, manage.
- **What is a PAN? Example?** [B · B] Personal Area Network around one person, about **10 m**: Bluetooth, USB, NFC. Example: phone ↔ smartwatch or earbuds. Then LAN (building) → MAN (city) → WAN (country).

4 Transmission Media & Ethernet



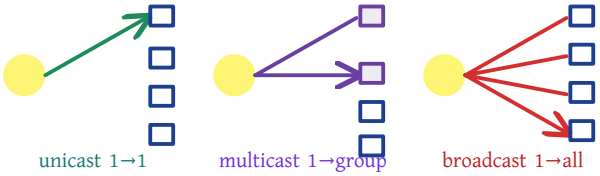
- **Optical fibre? Its principle?** [P] A thin glass/plastic strand carrying light. Core index > cladding index, so light beyond the critical angle undergoes **total internal reflection** and zig-zags along. Single- or multimode; no EMI.
- **How does the light become bits?** [P] A **photodetector** (PIN photodiode/APD) in the receiver/SFP turns light pulses into current → bits.
- **How do you detect a cut cable?** [P] Link LED off; a **cable tester** checks each pair; a TDR or fibre **OTDR** finds the break.
- **10Base-T? Maximum length?** [B · B] **10** Mbps Ethernet, **Base**band, **T**wisted pair (UTP, RJ-45, star, IEEE 802.3i). Segment limit **100 m.**
- **Synchronous vs asynchronous?** [N23] Async: each character framed by a **start bit and stop bit**, gaps allowed, no shared clock (UART, RS-232); ≥ 20% overhead. Sync: continuous blocks locked to **one clock**, sent on a line or recovered from the data (preamble, Manchester); no per-byte framing; Ethernet, HDLC.

Exam trap: The **modulator** (laser/LED driver) is at the transmitter: electrical → light. Light → bits is the **photodetector**. 10Base2 = 185 m, 10Base5 = 500 m.

5 IP & MAC Addressing, Casting

	IPv4 [B ×3]	IPv6
Length · notation	**32 bit** · 192.168.1.1	**128 bit** · 2001:db8::1 (hex)
Addresses · header	≈ 4.3 billion · 20–60 B	≈ 3.4×10^{38} · fixed 40 B
Config · casting	manual/DHCP · broadcast	SLAAC/DHCPv6 · **no broadcast**

- **IPv4? IPv6?** [B · B] Logical (L3) interface addresses. IPv4 private: 10/8, 172.16/12, 192.168/16. IPv6: 8 hex groups; one run of zero groups may be written **::**, once.
- **Static vs dynamic IP?** [B] Static: set by hand, fixed (servers, printers, ATMs). Dynamic: leased by DHCP, may change.
- **MAC address?** [B] **48-bit** NIC address (00:1A:2B:3C:4D:5E), first 24 bits = vendor; used inside the LAN; ARP maps IP → MAC.
- **Subnetting?** [B] Borrowing host bits to split a network: smaller broadcast domains, security, less waste. 192.168.10.0/24 → /26 = 4 subnets, each $2^6 - 2 = 62$ hosts.
- **Multicast vs broadcast?** [B ×3] Broadcast: one → **all** in the segment (255.255.255.255; ARP, DHCP Discover), stopped by routers. Multicast: one → a **subscribed group** (224–239.x.x.x, IGMP; IPTV, OSPF), saves bandwidth.



Exam trap: At each router hop the **MAC header is rewritten**; the IP addresses stay end to end. IPv6 has no broadcast.

6 PING & Troubleshooting

- **PING? Full form?** [B · B] **Packet InterNet Groper** (a backronym; named after sonar). ICMP Echo Request (type 8) → Echo Reply (type 0): reachability, round-trip time, TTL, loss.
- **Data-centre monitoring tools?** [B] Nagios, Zabbix, PRTG, SolarWinds; they poll by **SNMP** (UDP 161/162), NetFlow, syslog. Wireshark captures packets.

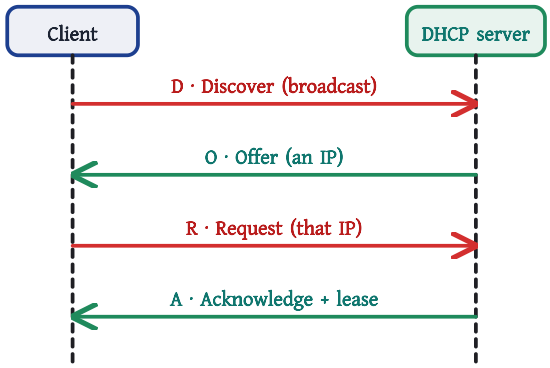
Model answer ✍: Branch PC has no Internet [B ×2]

1. Cable/Wi-Fi, link LED, other port.
2. *ipconfig /all*: 169.254.x.x = DHCP failed → renew; check gateway, DNS.
3. Ping 127.0.0.1 → gateway → 8.8.8.8 → a name (only names fail = DNS).
4. *tracert*; whole branch down → router/WAN/ISP: call the NOC.

7 Application Protocols & Ports

Port	Protocol	Port	Protocol	Port	Protocol
20/21	FTP data/control	53	DNS (UDP)	143 / 993	IMAP / IMAPS
22	SSH, SFTP	67/68	DHCP server/client	161/162	SNMP / trap
23	Telnet	**80**	HTTP	**443**	HTTPS
25	SMTP relay	110 / 995	POP3 / POP3S	3389	RDP

- **HTTP vs HTTPS? What is HTTPS?** [N23 · B] HTTP is plain text (port 80). **HTTPS = HTTP inside TLS** (formerly SSL), port 443: encryption, integrity, and server authentication by a CA-signed certificate.
- **Is HTTPS a protocol? Which layer?** [B · B] Yes, an application-layer (L7) protocol: HTTP over a TLS session. TLS sits between the application and TCP.
- **HTTP's port?** [N23] TCP **80** (8080 as an alternative).
- **GET vs POST?** [B] GET reads: parameters in the URL, kept in history/logs, cacheable, length-limited, idempotent. POST sends data in the **request body** (forms, uploads), not cached, no practical size limit, not idempotent.
- **What is DNS?** [B] Domain Name System: a hierarchical, distributed directory (root → TLD such as .bd → authoritative server) that resolves names to IPs. Records A, AAAA, MX, CNAME, PTR. UDP **53**.
- **DHCP? How does it work?** [B · B] Dynamic Host Configuration Protocol gives IP, mask, gateway, DNS automatically by **DORA**, UDP 67/68.
- **DHCP lease time?** [B] How long the IP is loaned (1 day on many routers, 8 days by default on Windows Server). Renew at 50%, rebind at 87.5%.
- **SMTP? Its ports?** [B · B] Simple Mail Transfer Protocol **pushes** mail (client → server, server → server). **25** relay, **587** submission + STARTTLS, **465** implicit TLS.



DORA: UDP 68 ↔ 67

Exam trap: POST is **not** "secure": without HTTPS its body is plain text on the wire. It only keeps data out of the URL, history and logs. Security comes from **TLS**, for GET and POST alike.

Memory trick: **SMTP sends, POP/IMAP pick up.** Ladder: 21 FTP · 22 SSH · 23 Telnet · 25 SMTP · 53 DNS · 67/68 DHCP · 80 HTTP · 110 POP3 · 143 IMAP · 443 HTTPS.

8 VLAN, VPN & Active Directory

- **VLAN? Why use it?** [B · B] A logical L2 segment made on switches by **IEEE 802.1Q** tags (4094
- **VPN? How does it work? Tunnelling?** [B ×3] An **encrypted tunnel over a public network**: ends

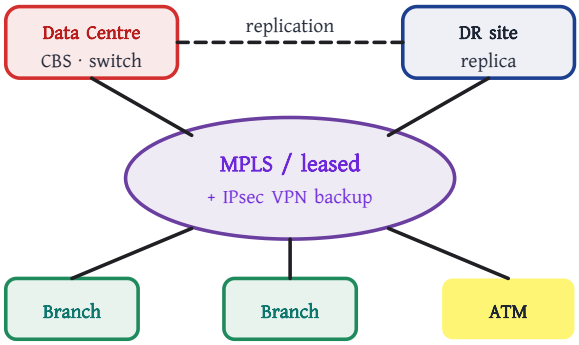
IDs); each is its own broadcast domain: **security** (ATM, CBS, staff, guest kept apart), less broadcast, grouping by function. Inter-VLAN traffic needs a router/L3 switch.

- **Active Directory? Domain controller?** [B · B] AD: Microsoft's LDAP directory of users, computers, groups and policies; single sign-on via Kerberos, Group Policy. A DC is the server running AD DS that authenticates log-ons (keep ≥ 2).

authenticate (IKE, certificates or pre-shared key), agree keys, then each packet is encrypted and **encapsulated** in a new IP packet and decrypted at the far end. IPsec (site-to-site), SSL VPN (remote users).

- **VPN vs VLAN?** [B] VPN securely joins **remote** sites over the Internet (encrypted, L3+). VLAN splits one **local** switched network (L2 tags, no encryption).

9 Bank WAN, Routing & ATM

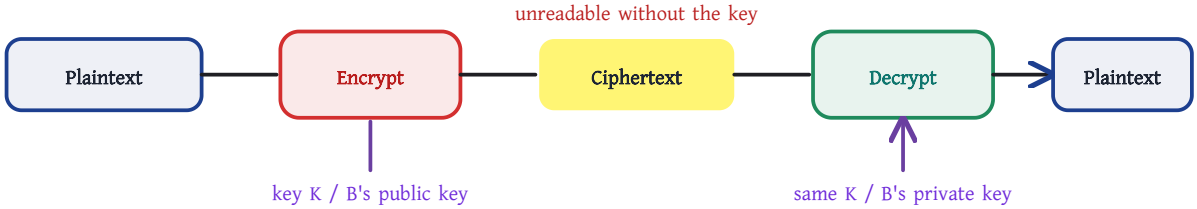


hub-and-spoke bank WAN

- **How are head office and branches (all branches) connected?** [B · B] A **hub-and-spoke WAN**: data centre + DR as the hub, branches over **MPLS VPN** or leased lines; backup by radio/4G or Internet IPsec VPN.
- **Physical media?** [B] Optical fibre first; radio/microwave, VSAT or cellular as backup.
- **Protocols between branches?** [B] TCP/IP over MPLS; IPsec/GRE tunnels; OSPF/EIGRP inside, BGP with carriers.
- **Does Sonali Bank use an intranet?** [B] Yes: a private TCP/IP network for CBS, mail, HR and portals, staff only.
- **Routing protocol? Name some.** [B · B] Rules for routers to swap routes and pick the best path. **RIP**: distance vector, hop count, max **15**. **OSPF**: link state, Dijkstra. **EIGRP**: Cisco hybrid. **IS-IS**. **BGP**: path vector between autonomous systems (the Internet).

> Model answer ✍️: **How does an ATM work in a network?** [B] Card + PIN (encrypted in the keypad) → **ISO 8583** request over the bank's private link → **card switch** → CBS checks and debits → approval → cash. Other banks' cards via **NPSB** or VISA/Mastercard.

10 Cryptography & Encryption



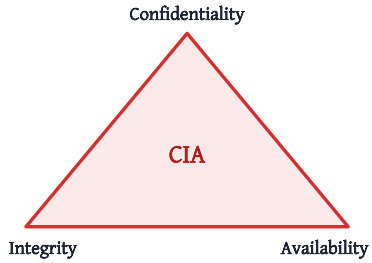
- **What is cryptography?** [P · N23] Securing data so only intended parties can read or verify it. Types: **symmetric** (one shared key), **asymmetric** (public/private pair), **hash** (one-way digest).
- **Data encryption? Plaintext vs ciphertext?** [B ×4] Encryption turns readable **plaintext** into unreadable **ciphertext** with an algorithm + key; only the right key decrypts it.

Algorithm	Type · key
AES ★	sym · 128/192/256
DES / 3DES	sym · **56** / 168
RSA / ECC	asym · 2048+ / 256
SHA-256	hash · 256-bit digest

- **Algorithms and tools?** [B] Table; tools: BitLocker, VeraCrypt, OpenSSL, GnuPG.
- **Blockchain?** [P] A shared append-only ledger; each block holds the **previous block's hash**, so tampering breaks the chain.

> Model answer ✍️: **Secure data exchange in an Android app; what do developers gain?** HTTPS/TLS only, certificate pinning · token auth (OAuth 2.0/JWT) checked on **every** API call · data at rest encrypted via the Android Keystore · runtime permissions (since 6.0), least privilege · no secrets in logs. Gain: fewer breaches, less liability, user trust, compliance.

11 Security Principles & Authentication

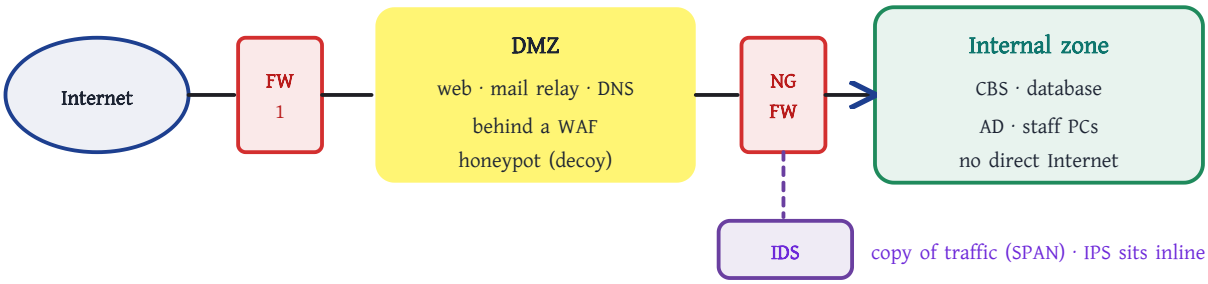


- **Basic principles? CIA triad?** [B · B] **Confidentiality** (encryption, access control), **Integrity** (hashes, signatures), **Availability** (redundancy, backups, DR); plus authentication, authorisation, accounting, **non-repudiation**.
- **2FA? MFA?** [B · B] Factors: something you **know** (PIN), **have** (OTP phone, card), **are** (fingerprint). 2FA = two *different* factors (card + PIN); MFA = two or more.
- **Host security? Data security?** [B · B] Host: hardening each endpoint (patches, EDR, host firewall). Data: protecting data at rest, in transit, in use (encryption, DLP, backups).

> Memory trick: **S**ymmetric = **S**ame key; **A**symmetric = **A** pair; hash = fingerprint.

> Exam trap: Hashing, Base64 ≠ encryption. Password + PIN ≠ 2FA.

12 Firewalls, IDS/IPS, DMZ & Honeypots



defence in depth: a DMZ between two firewalls

- **What is a firewall?** [B] Hardware or software that **filters traffic between trust zones** by rules (IP, port, protocol, direction), default deny.
- **NGFW?** [B] Next-Generation Firewall: stateful firewall + **application awareness** (deep packet inspection), built-in IPS, user identity, TLS inspection, URL filtering.
- **WAF?** [B] Web Application Firewall: an **L7** filter in front of web apps that blocks **SQL injection, XSS**, CSRF and similar.
- **NGFW vs WAF?** [B] NGFW guards the whole network (all ports, apps); WAF guards web apps only, but reads HTTP in depth. Banks run both.
- **IDS? IPS? Difference?** [B ×3] **IDS** watches a copy of traffic and **alerts** (passive). **IPS** sits **inline** and drops bad traffic in real time; a false positive blocks real users. Snort, Suricata.
- **DMZ? DMZ vs internal zone?** [B · B] A buffer subnet for **public-facing servers** (web, mail relay, Internet-banking front end). If one is hacked, the internal zone (CBS, database, AD: private, most trusted) is still behind a second firewall.
- **Honeypot?** [B] A decoy with no real use: any traffic to it is suspicious, so it warns early and records attackers' methods.

> Model answer ✍️: **How does a firewall work?** [B] Each packet is checked against an ordered rule list (ACL): **first match wins**, implicit "deny all" at the end. Packet filter: L3/L4 headers only. **Stateful**: tracks connections, lets replies back in. Proxy: inspects L7 content. NGFW: all of these + DPI + IPS.

> Memory trick: **IDS = Detect & Shout** (CCTV). **IPS = Prevent & Stop** (guard at the gate).

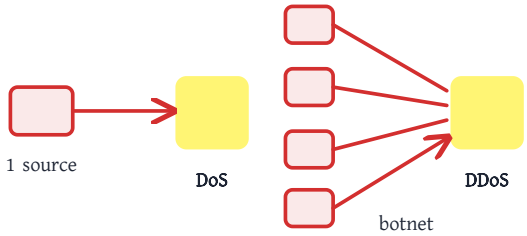
> Exam trap: A firewall does not stop phishing or an insider. Never put the CBS database in the DMZ.

13 Phishing, Malware, DoS & Ethical Hacking

- **What is phishing?** [B] A **social-engineering** attack: a fake e-mail, SMS or site posing as a bank or wallet to steal passwords, OTPs, card data. Spear phishing, smishing (SMS), vishing (voice).
- **How do you identify it?** [B] Look-alike domain, link ≠ text (hover), urgency, a request for PIN/OTP (banks never ask), spelling errors, odd attachments. Verify by the official channel.
- **Account hacked: phishing?** [B] Not necessarily: weak/reused passwords, malware, SIM swap, session hijack too.
- **Is phishing ethical?** [B] Real phishing is a crime; an **authorised** staff-training simulation is ethical.

Malware [B ×2]	Key trait
Virus	rides a file, needs a user
Worm	self-spreads, no host
Trojan	looks legit, back door
Ransomware	encrypts data for ransom
Spyware, keylogger	secretly records
Rootkit, adware	hides in OS · ads

- **Antivirus? Examples?** [B · B] Detects (signatures, heuristics, behaviour) and removes malware: Defender, Kaspersky, Bitdefender, ESET, Norton.



- **DoS? DDoS? Difference?** [B ×3] **Denial of Service**: one source floods a target to exhaust bandwidth, CPU or memory (SYN flood). **Distributed**: a botnet attacks at once, so blocking one IP fails.
- **Prevention?** [B] Firewall/ACLs, rate limiting, SYN cookies, IPS, load balancer/CDN, ISP or cloud scrubbing, monitoring.
- **Penetration testing?** [B] An **authorised**, scoped, simulated attack: recon → scan (Nmap) → exploit (Metasploit) → report with fixes.
- **Hacking vs ethical hacking?** [B] The difference is **permission**: a white hat works under written authorisation and reports every finding.

Memory trick: **Virus rides, worm walks, Trojan disguises, ransomware locks.**

Exam trap: The padlock (HTTPS) proves an encrypted link, **not** a genuine site: phishing sites have certificates too.

Night-Before-the-Viva Sheet

Fact sheet:

OSI 7: Ph · DL · N · T · S · P · A

TCP/IP 4: App · Transport · Internet · Net access

Hub/repeater L1 · switch L2 · router L3

Gateway = protocol translation, any layer

Framing = data link layer (FCS/CRC)

10Base-T: 10 Mbps, UTP, 100 m

IPv4 32 bit · IPv6 128 bit · MAC 48 bit

Multicast 224–239.x.x.x · IPv6: no broadcast

HTTP 80 · HTTPS 443 · DNS 53 · SSH 22

SMTP 25 / 587 / 465 · DHCP UDP 67/68

DHCP = DORA · renew at 50% of lease

PING = ICMP echo 8 / reply 0

VLAN = 802.1Q · VPN = encrypted tunnel

RIP max 15 hops · OSPF link state · BGP

AES 128-bit block · DES 56-bit key

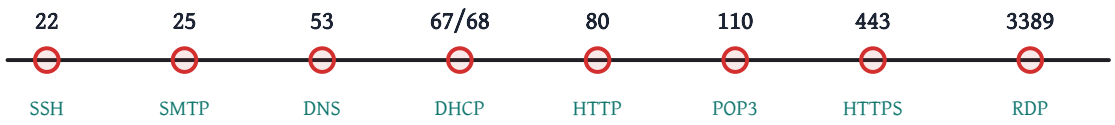
CIA + AAA + non-repudiation

Top 12 one-liners for the board

1. OSI is ISO's reference model, not a protocol
2. Switch forwards by MAC, router by IP
3. Default gateway = your router's interface
4. Fibre works by total internal reflection
5. Photodiode turns light into bits
6. Star topology rules offices; PAN ≈ 10 m
7. HTTPS = HTTP + TLS, application layer
8. VLAN separates inside, VPN connects outside
9. IDS alerts; IPS blocks inline
10. DMZ hosts public servers, never the CBS
11. DDoS = many sources (botnet)
12. 2FA = two different factor types

Top traps: optical *modulator* ≠ receiver (photodetector) · POST is not secure without HTTPS · TCP/IP's last layer is network *access* · "layers 1–3 = hardware" is loose · a router alone serves a few PCs (SOHO router has a switch) · padlock ≠ genuine site · hashing ≠ encryption

All tricks together: Please Do Not Throw Sausage Pizza Away · ATIN · hub shouts, switch reads the MAC, router reads the IP · DORA · SMTP sends, POP/IMAP pick up · Symmetric = Same key · IDS shouts, IPS stops · troubleshoot bottom-up · virus rides, worm walks



the port ruler: say it aloud once before you sleep

✓ Practice MCQs (exam pattern, 10 marks)

1. A standard switch works at OSI layer:
   (a) Physical (b) Network
   (c) Data link (d) Transport

2. Which one regenerates a weak signal?
   (a) Repeater (b) Router
   (c) Bridge (d) Gateway

3. Mail submission with STARTTLS uses port:

(a) 25 (b) 465
(c) 110 (d) 587

4. Maximum segment length of 10Base-T:

(a) 185 m (b) 100 m
(c) 500 m (d) 2 km

5. Fibre receiver: light → current in a:

(a) laser diode (b) modulator
(c) photodiode (d) LED

6. The first DHCP message a client sends:

(a) Discover (b) Request
(c) Offer (d) Ack

7. Which one blocks attacks inline?

(a) IDS (b) Honeypot
(c) Syslog (d) IPS

8. Password + OTP on a phone is:

(a) one factor (b) 2FA
(c) biometric (d) encryption

9. An IPv4 multicast address:

(a) 230.1.1.1 (b) 10.1.1.1
(c) 127.0.0.1 (d) 169.254.1.1

10. A WAF mainly stops:

(a) cable cuts (b) ARP spoofing
(c) SQL injection (d) power loss

Answer key → 1-c · 2-a · 3-d · 4-b · 5-c · 6-a · 7-d · 8-b · 9-a · 10-c

Written question: Sketch a secure network for a bank's head office that also serves its branches and ATMs. Explain the role of the firewall, DMZ, IDS/IPS, VLANs and VPN. 10 marks

1. **Zones** (sketch as on p. 9): Internet → firewall → **DMZ** (web front end behind a WAF) → NGFW → **internal zone** (CBS, database, AD). 3
2. **Firewalls** enforce default-deny between zones; NGFW adds an inline **IPS**; an **IDS** alerts. 2
3. **VLANs** separate CBS servers, staff PCs, ATMs and guest Wi-Fi; ACLs between them. 2
4. **Branches and ATMs** reach the data centre over MPLS or IPsec **VPN**; DR site replicates. 2
5. AD with MFA for admins, logging, patching, backups: all three parts of CIA. 1

Before the viva ✓

- ✓ OSI 7 and TCP/IP 4, both directions
- ✓ Every device on its layer
- ✓ The 10 key port numbers
- ✓ DORA, VPN tunnel, firewall rules
- ✓ IDS/IPS, DoS/DDoS, NGFW/WAF

Say it aloud (2 min):

① What happens when you type a URL
② How a branch reaches the CBS
③ Stopping a phishing wave
④ Hub vs switch vs router